

09/02/2025

Finite Automaton

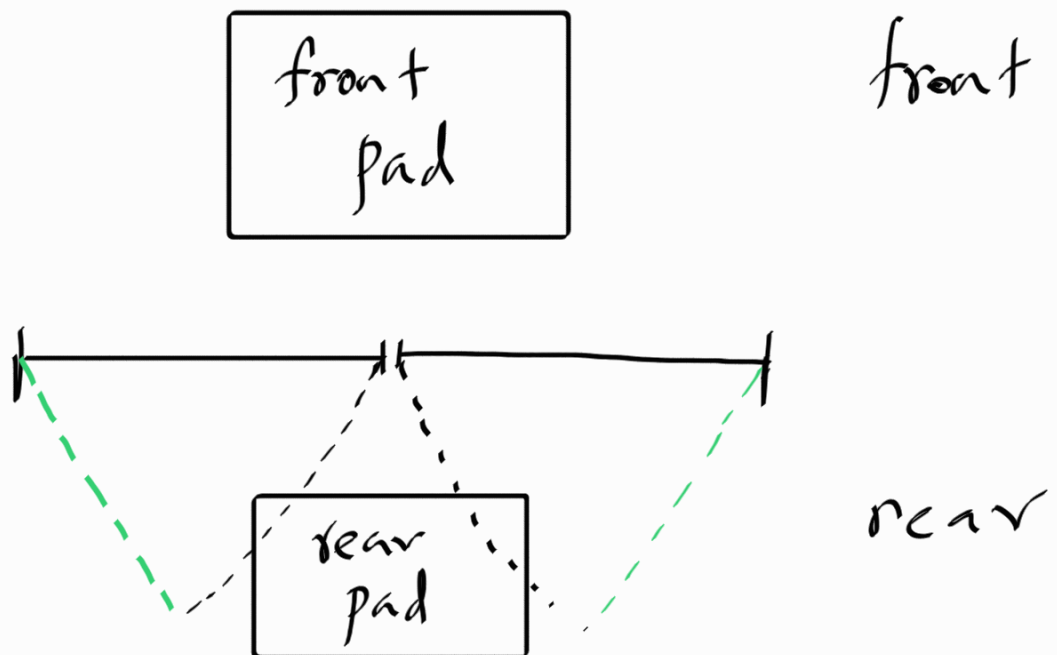
What is a computational model?

Computational model is an idealized version of a computer.

- computation models are accurate in some ways and not in others.

Finite Automata

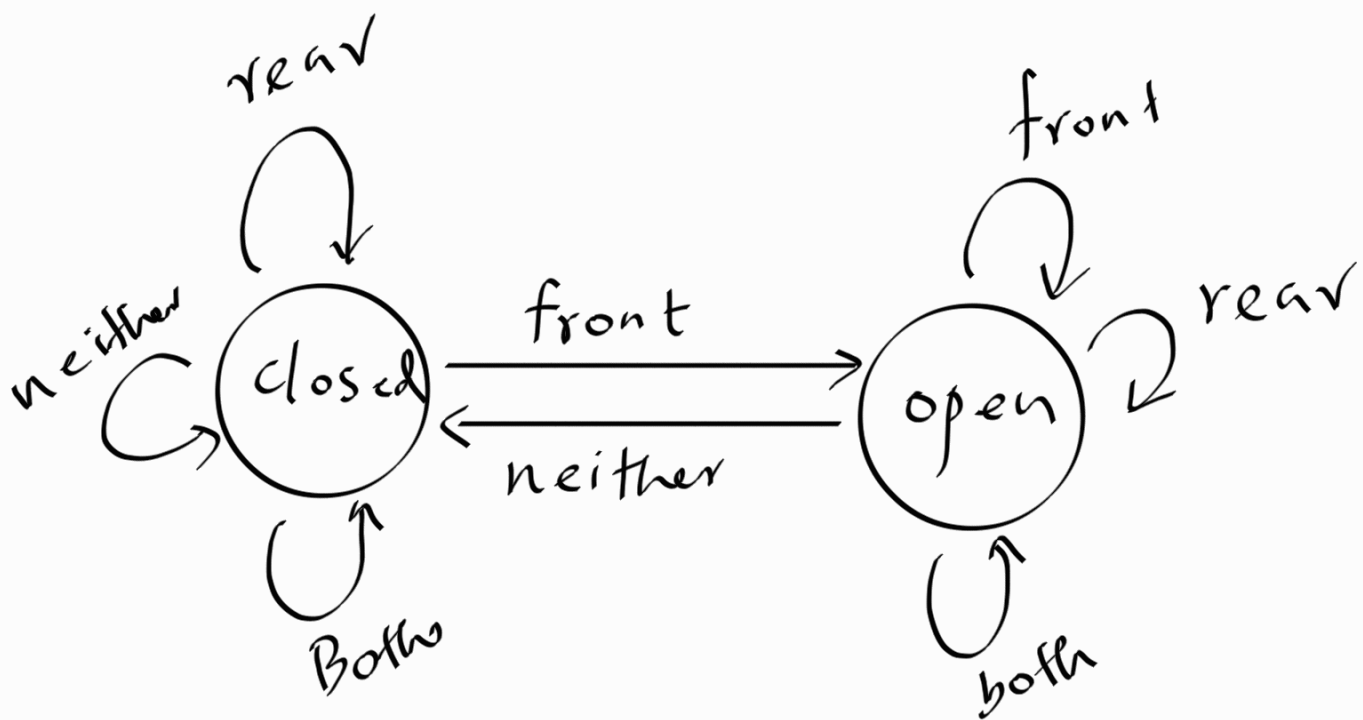
- A model to understand computers with extremely limited amount of memory.



controller is either closed or open.

There are 4 possible inputs

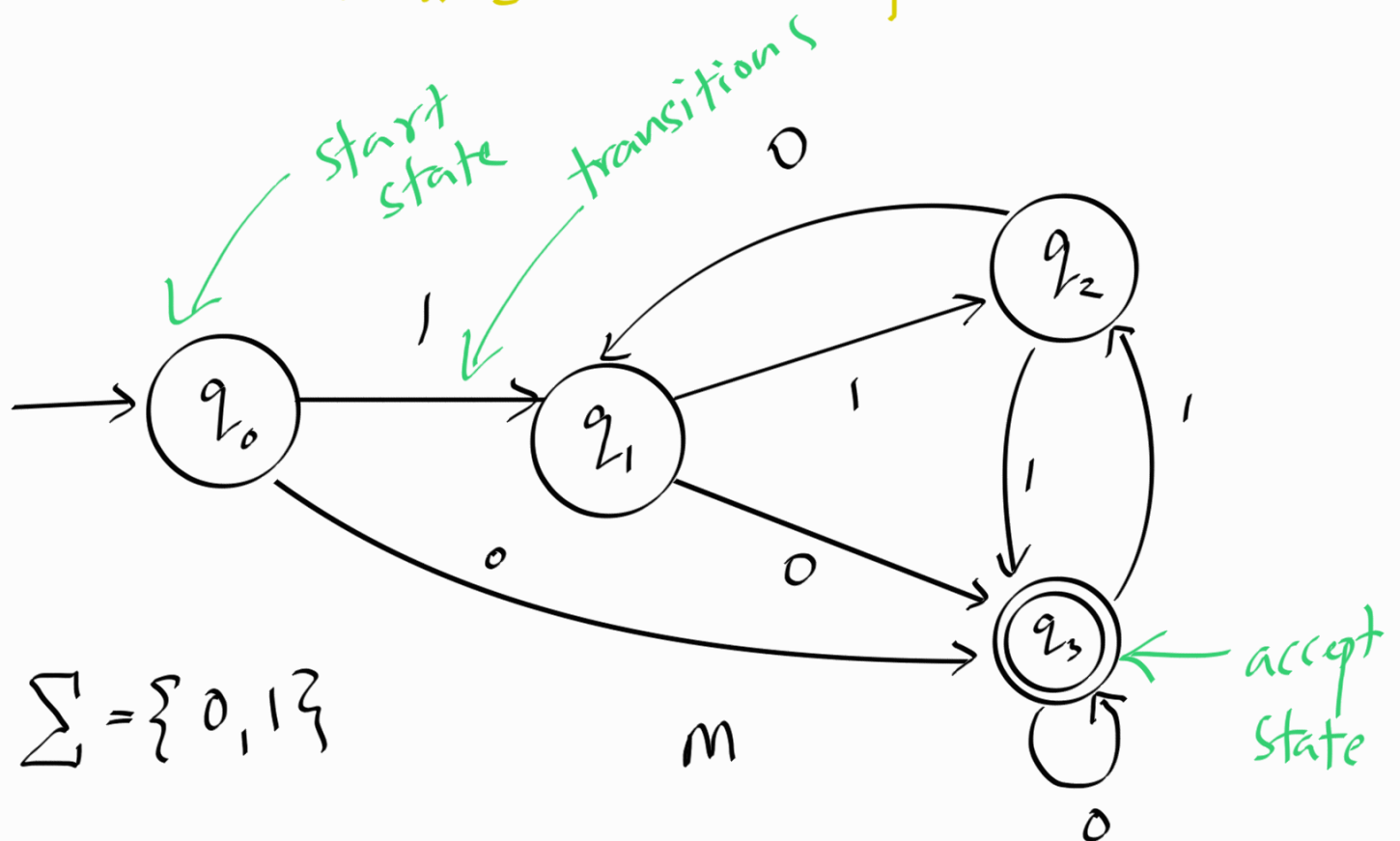
1. Front
2. Rear
3. Both
4. Neither



The memory of this computer is 1 bit.

Let's look at this in a mathematical perspective. Let's abstract this concept of finite automaton, without any reference to any application.

- Basic idea is that a finite automaton has finite set of states (things that the computer can keep track of).
- Depending on the current input machine changes the state.
- Machine must have a start state.
- Machine has a set of accept states.



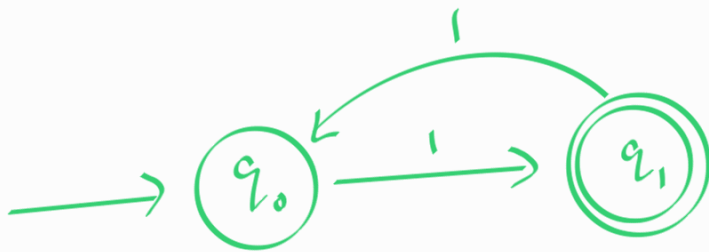
10 is
accepted by M

101 is not
accepted by M

Ex: Design a finite automaton which accepts

$A = \{w \mid \text{the length of } w \text{ is odd}\}$

$\Sigma = \{1\}$



111 ✓
11 ✗

state that represents
we have processed
odd # of 1's upto
now.

- States are nodes, some states are accepting states

- arrows represent transitions.

- inputs are symbols available in the alphabet.

Definition

- A finite state automaton is a 5-tuple: $(Q, \Sigma, \delta, q_0, F)$
 - Q is the set of states $Q = \{q_0, q_1\}$
 - Σ is the alphabet. $\Sigma = \{0, 1, 2\}$
 - $\delta : Q \times \Sigma \rightarrow Q$
 - $q_0 \in Q$ is the start state
 - $F \subseteq Q$ is the set of accepting states.
- $F = \{q_1\} \subseteq Q$

If A is the set of strings that is accepted by the machine M , then

$$L(M) = A$$

M recognizes A .

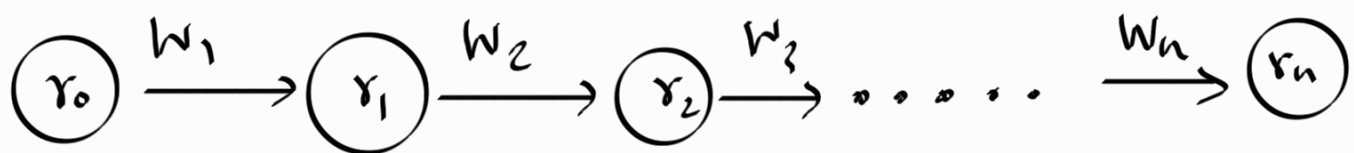
M accepts A .

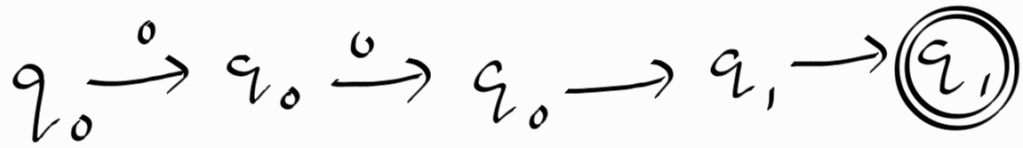
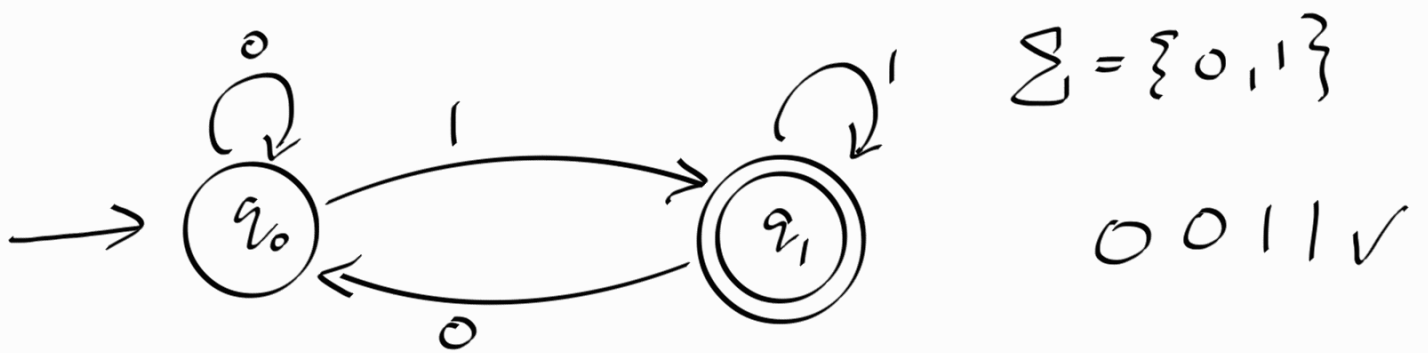
Formally, given $M = (Q, \Sigma, \delta, q_0, F)$ and $w = w_1 w_2 w_3 \dots w_n$ be a string over alphabet Σ , M accepts w if there is a sequence of states $r_0, r_1, r_2, \dots, r_n$ such that

1. $r_0 = q_0$

2. $\delta(r_i, w_{i+1}) = r_{i+1} \quad ; \forall i \in \{0, 1, 2, \dots, n-1\}$

3. $r_n \in F$

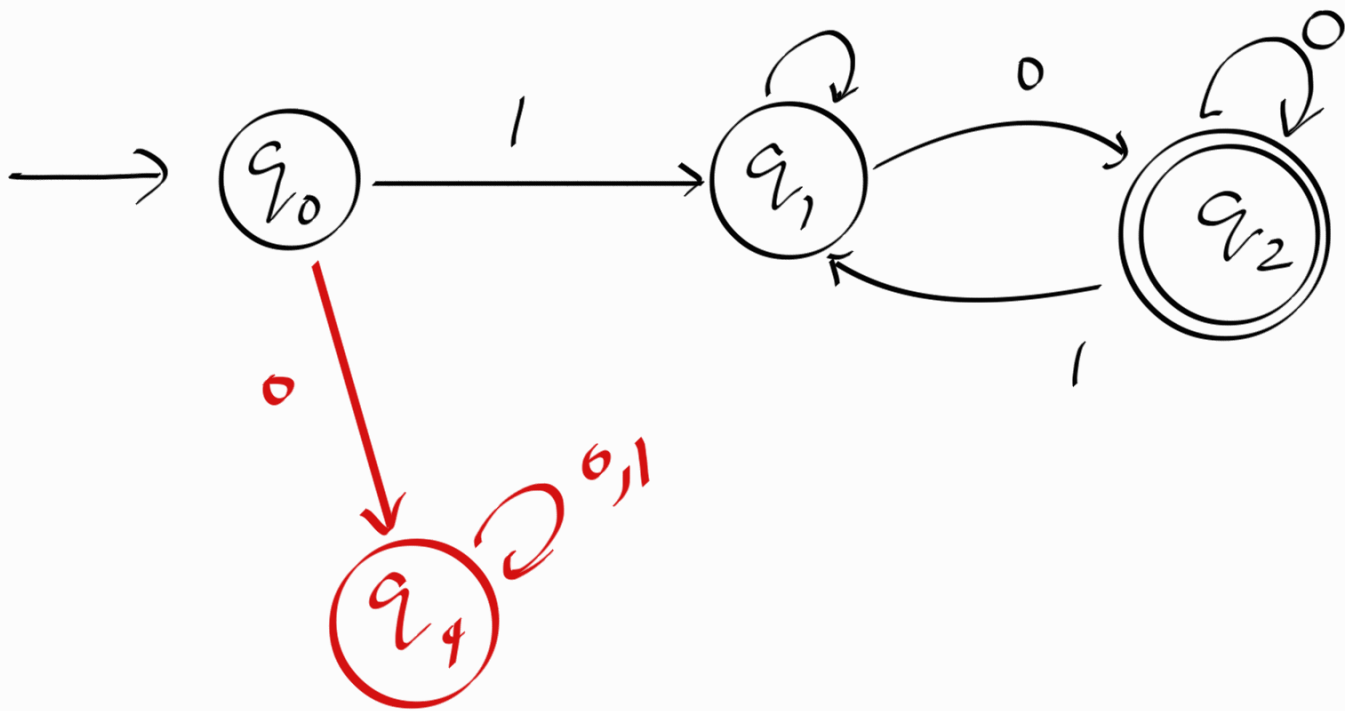




010 ✗



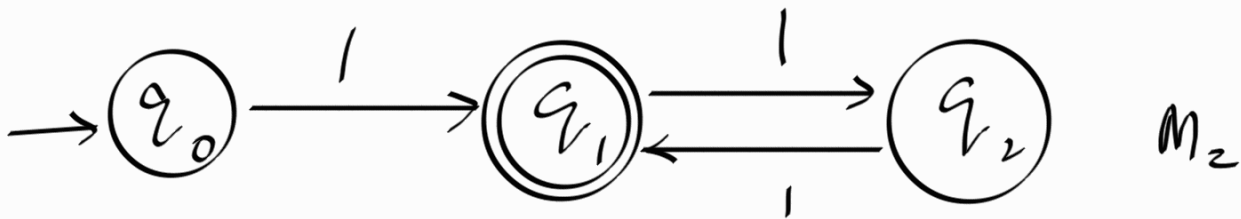
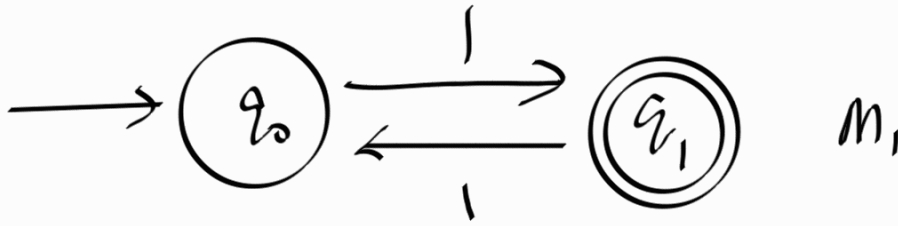
Suppose you have the following



Question! Is q_4 needed?

Yes, we need this formally but since the transition moves to a state that will not lead to an accept state, we could drop it.

$$\Sigma = \{1\}$$

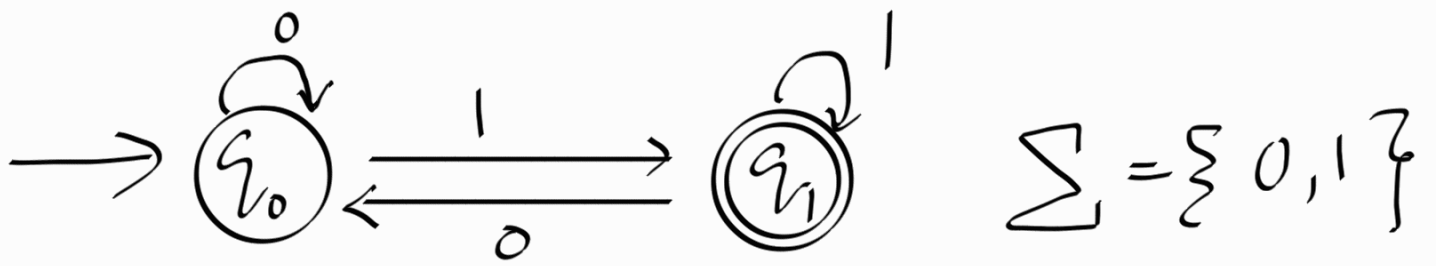


Do these machines accept the same language?

Yes

$$L(M_1) = L(M_2)$$

$$M_1 = M_2 \iff L(M_1) = L(M_2)$$



1. $Q = \{q_0, q_1\}$

2. $\Sigma = \{0, 1\}$

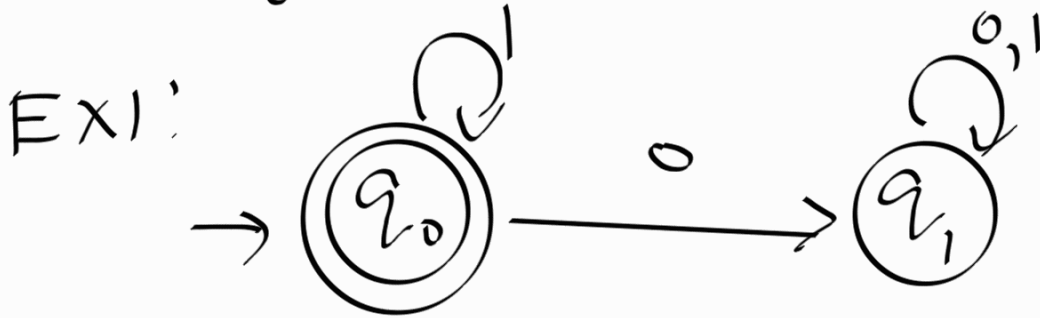
3.

	0	1
q_0	q_0	q_1
q_1	q_0	q_1

4. q_0 is the start state

5. $F = \{q_1\} \quad \{q_1\} \subseteq Q$

Let us try to identify
languages accepted by the
following machines. $\Sigma = \{0, 1\}$



$$L(M_1) = \{1^n \mid n \geq 0\}$$

$$L(M_1) = 1^*$$